

Academic Qualifications

Year	Qualification	Institute	CPI/%
2022 - Present	B.Tech.	Indian Institute of Technology Kanpur	8.3/10
2022	Class XII (CBSE)	Delhi Public School Nigahi	93.2%
2020	Class X (CBSE)	Delhi Public School Nigahi	95.2%

Scholastic Achievements

- Secured **Bronze medal** in **Inter-IIT Tech Meet 13.0 (2024)** for the Albatross Energetics’ HVAC problem statement
- Awarded **KVPY Fellowship** for qualifying **KVPY SA 2020** with **All India Rank 641** amongst **1 lakhs+** candidates
- Secured a **Top 100** rank in **InMobi Codies 2026**, a flagship national-level hackathon organized by InMobi among IITs

Professional Experience

Software Engineering Intern | Samsara

(May’26-Jul’26)

Objective	Redesigned a large-scale public API ’s pagination to fix unreliable behavior on large, sparse historical queries
Approach	- Authored RFCs benchmarking legacy, point-bounded & DuckDB-based designs to select the optimal approach - Engineered a reusable Go pagination engine as a point-counting wrapper over a byte-budgeted reader - Integrated pagination engine via gRPC across microservices with backward compatibility and staged rollout - Hardened system with table-driven, property-based, integration, and race testing across varying workloads
Outcome	Reduced p90 per-page latency by 91.1% versus legacy, validating zero errors across 200M+ data points

Key Projects

Counselling Management Platform | Prof. Sudhanshu Shekhar Singh, CMHW, IIT Kanpur

(Oct’25-Feb’26)

- Architected a **full-stack** counselling platform with **REST APIs** using **TypeScript, Next.js, NestJS, PostgreSQL, Prisma**
- Engineered **privacy-preserving** data access with **field-level encryption & HMAC-based blind indexing** for sensitive records
- Designed **concurrency-safe** appointment booking with **ACID transactions, row-level locking** and **idempotency safeguards**
- Built **multi-device authentication** with **JWT access/refresh token rotation**, HTTP-only cookies and **session revocation**
- Implemented **real-time WebSocket notifications** and validated workflows via **unit, integration** and **end-to-end testing**

Event-Driven Kinetic Monte Carlo Engine | Undergraduate Project | Prof. Salman A. Khan

(Jan’25-Nov’25)

- Built a high-throughput, event-driven **Kinetic Monte Carlo engine** for scalable, stochastic lattice simulations in Python
- Implemented a **Binary Sum Tree**, reducing event selection and rate-update complexity from **O(N)** to **O(log N)** per step
- JIT-compiled the simulation kernel with **Numba** and parallelized independent simulation runs across multiple **CPU cores**
- Benchmarked **serialization and I/O strategies** for output files across storage size, read/write throughput and memory overhead
- Improved the engine’s measured event-processing throughput from **10** to over **100,000 events/second**, a **10⁴-fold** speedup

Parallel Computing | CS633 Course Project | Prof. Preeti Malakar

 (Feb’26-Apr’26)

- Parallelized a **3D stencil computation** using **MPI**, flattening sub-domains into padded **1D arrays** for cache locality
- Engineered **non-blocking halo exchanges** via **async messaging** and **null-processor routing** to avoid distributed deadlocks
- Profiled **scalability up to 96 processes** on **PARAM Rudra**, identifying **network communication** as the primary bottleneck

Computer Graphics | CS360 Course Project | Prof. Soumya Dutta

 (Aug’25-Nov’25)

- Built a modular **WebGL2 rendering engine** with reusable **geometry, buffer, and scene-management** abstractions
- Implemented **GPU shader pipelines** in **GLSL** for **Phong shading, ray tracing**, and framebuffer-based **shadow mapping**
- Developed an interactive **real-time renderer** supporting dynamic lighting, reflections, shadows, textures, and camera animation

Multi-Robot Target Surveillance | Summer Research Project | Prof. Twinkle Tripathy

(May’25-Jul’25)

- Developed a **multi-robot surveillance system** in **ROS2** for simultaneous target tracking using **range-based control logic**
- Designed and tuned **PID controllers** for angular and linear motion, using **OptiTrack** for real-world position sensing
- Implemented custom **target-reassignment logic**, enabling a **generalized, reusable framework** for any robot count

Technical Skills

- Programming Languages & Utilities:** C, C++, Go, Python, TypeScript, JavaScript, SQL, Bash, GLSL, L^AT_EX
- Software & Libraries:** Git, GitHub, Docker, Postman, gRPC, MPI, ROS2, Next.js, NestJS, PostgreSQL, Prisma, Numba

Relevant Coursework

* - Ongoing

Data Structures and Algorithms	Introduction to Machine Learning	Fundamentals of Computing
Digital Electronics	Introduction to Computer Graphics	Signals, Systems & Networks
Parallel Computing	Computer Arithmetic on FPGA*	Convex Optimization in ML

Leadership Experience

Coordinator | Center for Mental Health and Wellbeing, IIT Kanpur

(Apr’25-Apr’26)

- Spearheaded **3-tier** team, overseeing **22** Core Team Members, **350+** Guides/Mentors; led **9-day** Orientation for **1200+** students
- Managed a **INR 40L+** budget and secured **INR 10L** in sponsorships through strategic vendor and bank partnerships
- Special Invitee to **Senate, IITK**, institute’s highest decision & policy making body and **Student Wellness Committee**, IITK
- Revamped the **CMHW website** (**183%** engagement increase) and expanded **online counselling** access, driving a **145%** rise